

ROLE OF MUDA, MURI & MURA IN IMPLEMENTING HEIJUNKA



MUDA – NON-VALUE ADDED

JAPANESE TERM OF NON-VALUE ADDED WASTE

IT IS ALSO ONE OF THE 8 LEAN WASTES

**WASTEFUL ACTIVITIES THAT INCREASE CYCLE TIME, CAUSE
ADDITIONAL MOVEMENTS TO GET PARTS OR TOOLS, CREATE
EXCESS INVENTORY, OR RESULT IN ANY TYPE OF WAITING**

MURI – OVERBURDENING OF PEOPLE OR EQUIPMENT

PUSHING AN INDIVIDUAL OR AN EQUIPMENT BEYOND NATURAL LIMITS

RESULTS IN SAFETY RELATED ISSUES

ALSO INCREASES THE PROBABILITY OF CAUSING ERRORS

OVERBURDENING EQUIPMENT CAUSES BREAKDOWNS AND DEFECTS

MURA – UNEVENNESS

ON THE SHOP FLOOR, WORK IS UNEVEN

**IT IS A RESULT OF AN IRREGULAR PRODUCTION SCHEDULE OR
FLUCTUATING PRODUCTION VOLUMES**

**OTHER REASONS – DOWNTIME, MISSING PARTS, MISSING
INFORMATION, ETC**

MUDA IS A CONSEQUENTIAL RESULT OF UNEVENNESS OR MURA

**TO HANDLE THIS, STAFFING AND INVENTORY LEVELS ARE ALWAYS
KEPT AT LEVELS THAT CAN HANDLE PEAK VOLUMES EVEN IF THE
AVERAGE REQUIREMENTS ARE MUCH LOWER THAN THAT**

HOW MUDA, MURI & MURA CAN WORK TOGETHER?

LET'S SAY YOU HAVE A TROUBLED PRODUCTION SCHEDULE

**FIRST, LOOK FOR WAYS TO
REDUCE INVENTORY LEVELS
FROM YOUR PRODUCTION
PROCESS**

**SECOND, LOOK AT THE
WORK BALANCE AND
REDUCE THE NUMBER OF
PEOPLE OR EXCESS CAPACITY**

**THIRD, ORGANIZE THE
WORKPLACE BETTER TO
REDUCE WASTED MOTION**

SYSTEM WILL START FACING TURBULENT TIMES

IT WILL FORCE PEOPLE AND EQUIPMENT TO WORK HARDER

**WORKERS WILL BE OVERBURDENED, EQUIPMENT WILL BREAK-
DOWN, YOU WILL RUN OUT OF PARTS...**

LEAN MANAGEMENT DOESN'T WORK!

HOW MUDA, MURI & MURA CAN WORK TOGETHER?

WHAT MANY COMPANIES FAIL TO DO WHILE IMPLEMENTING LEAN IS STABILIZING THE SYSTEM AND CREATING “EVENNESS” – A TRUE BALANCE OF LEAN FLOW OF WORK

THIS IS HEIJUNKA – LEVELING OUT THE WORK SCHEDULE

ACHIEVING HEIJUNKA IS FUNDAMENTAL TO ELIMINATING MURA (OR UNEVENNESS), WHICH IS FUNDAMENTAL TO ELIMINATING MURI (OR OVERBURDEN) AND MUDA (WASTE)